In [18]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [19]:
```python
titanic_dataset = sns.load_dataset("titanic")
print(titanic_dataset.head(10))
print(titanic_dataset.info(10))
print(titanic_dataset.describe())
```

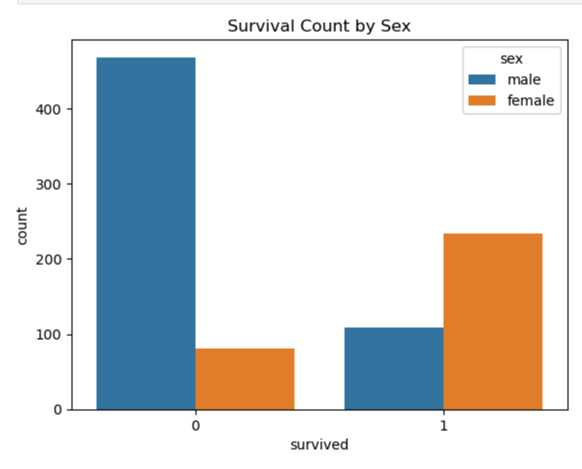
```
   survived  pclass     sex   age  sibsp  parch     fare embarked   class  \
0         0       3    male  22.0      1      0   7.2500        S   Third
1         1       1  female  38.0      1      0  71.2833        C   First
2         1       3  female  26.0      0      0   7.9250        S   Third
3         1       1  female  35.0      1      0  53.1000        S   First
4         0       3    male  35.0      0      0   8.0500        S   Third
5         0       3    male   NaN      0      0   8.4583        Q   Third
6         0       1    male  54.0      0      0  51.8625        S   First
7         0       3    male   2.0      3      1  21.0750        S   Third
8         1       3  female  27.0      0      2  11.1333        S   Third
9         1       2  female  14.0      1      0  30.0708        C  Second

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True
5    man        True  NaN   Queenstown    no   True
6    man        True    E  Southampton    no   True
7  child       False  NaN  Southampton    no  False
8  woman       False  NaN  Southampton   yes  False
9  child       False  NaN    Cherbourg   yes  False
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None
         survived      pclass         age       sibsp       parch        fare
count  891.000000  891.000000  714.000000  891.000000  891.000000  891.000000
mean     0.383838    2.308642   29.699118    0.523008    0.381594   32.204208
std      0.486592    0.836071   14.526497    1.102743    0.806057   49.693429
min      0.000000    1.000000    0.420000    0.000000    0.000000    0.000000
25%      0.000000    2.000000   20.125000    0.000000    0.000000    7.910400
50%      0.000000    3.000000   28.000000    0.000000    0.000000   14.454200
75%      1.000000    3.000000   38.000000    1.000000    0.000000   31.000000
max      1.000000    3.000000   80.000000    8.000000    6.000000  512.329200
```

In [20]:
```python
clean_age = titanic_dataset['age'].dropna()
clean_deck = titanic_dataset['deck'].dropna()
```

In [21]:
```python
total_passengers = titanic_dataset['survived'].count()
survived_passengers = titanic_dataset['survived'].sum()
survival_rate = survived_passengers / total_passengers * 100
print(f"Total Passengers: {total_passengers}")
print(f"Survived Passengers: {survived_passengers}")
print(f"Survival Rate: {survival_rate:.2f}%")
```

```
Total Passengers: 891
Survived Passengers: 342
Survival Rate: 38.38%
```

In [22]:
```python
sns.countplot(titanic_dataset, x='survived', hue='sex')
plt.title('Survival Count by Sex')
plt.show()
```



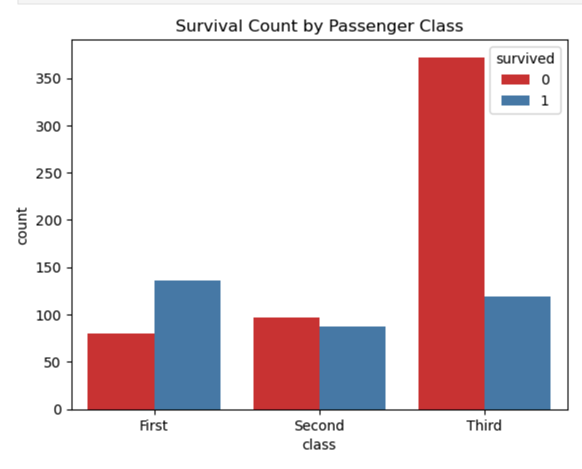
In [23]:
```python
titanic_dataset['sex'].value_counts()
```

Out[23]:
```
sex
male      577
female    314
Name: count, dtype: int64
```

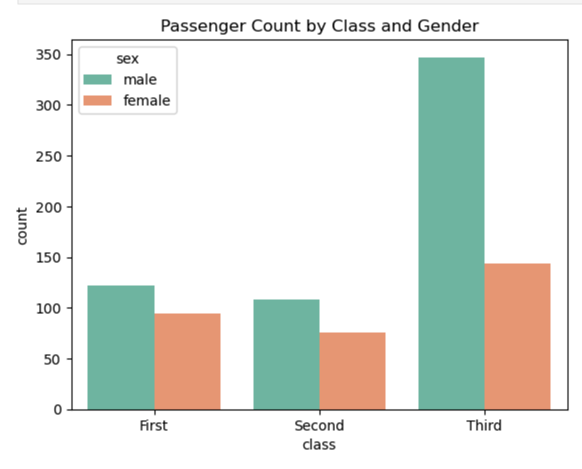
In [24]:
```python
titanic_dataset.groupby('sex')['survived'].mean()*100
```

Out[24]:
```
sex
female    74.203822
male      18.890815
Name: survived, dtype: float64
```

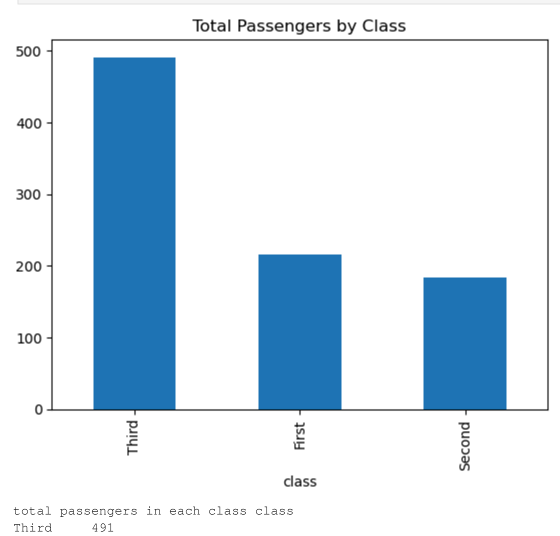
In [25]:
```python
sns.countplot(titanic_dataset, x='class', hue='survived', palette='Set1')
plt.title('Survival Count by Passenger Class')
plt.show()
```



In [26]:
```python
sns.countplot(titanic_dataset, x='class', hue='sex', palette='Set2')
plt.title('Passenger Count by Class and Gender')
plt.show()
```



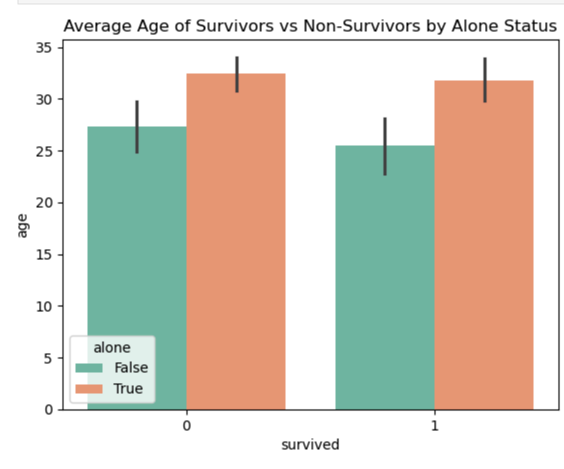
In [27]:
```python
titanic_dataset.value_counts('class').plot(kind='bar')
plt.title('Total Passengers by Class')
plt.show()

print('total passengers in each class',titanic_dataset['class'].value_counts())
titanic_dataset.groupby('class', observed=True)['survived'].mean()*100
```



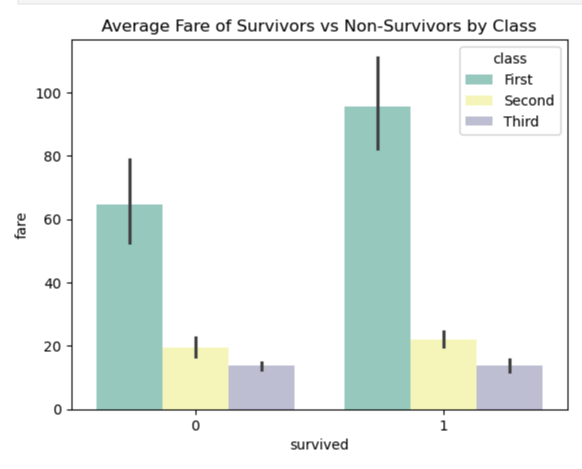

```
total passengers in each class class
Third     491
First     216
Second    184
Name: count, dtype: int64
```

Out[27]:
```
class
First     62.962963
Second    47.282609
Third     24.236253
Name: survived, dtype: float64
```

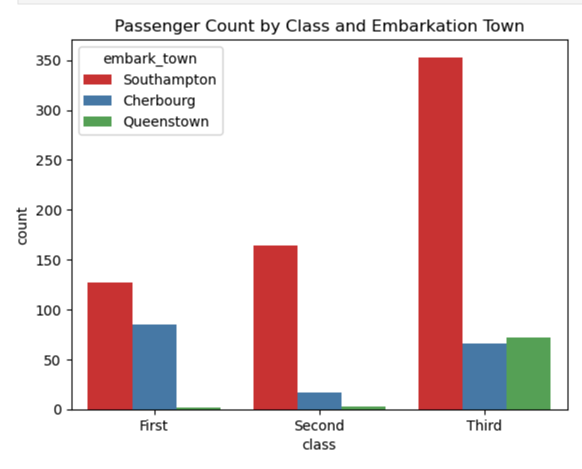
In [28]:
```python
sns.barplot(titanic_dataset, x='survived', y='age', hue='alone', palette='Set2')
plt.title('Average Age of Survivors vs Non-Survivors by Alone Status')
plt.show()
```



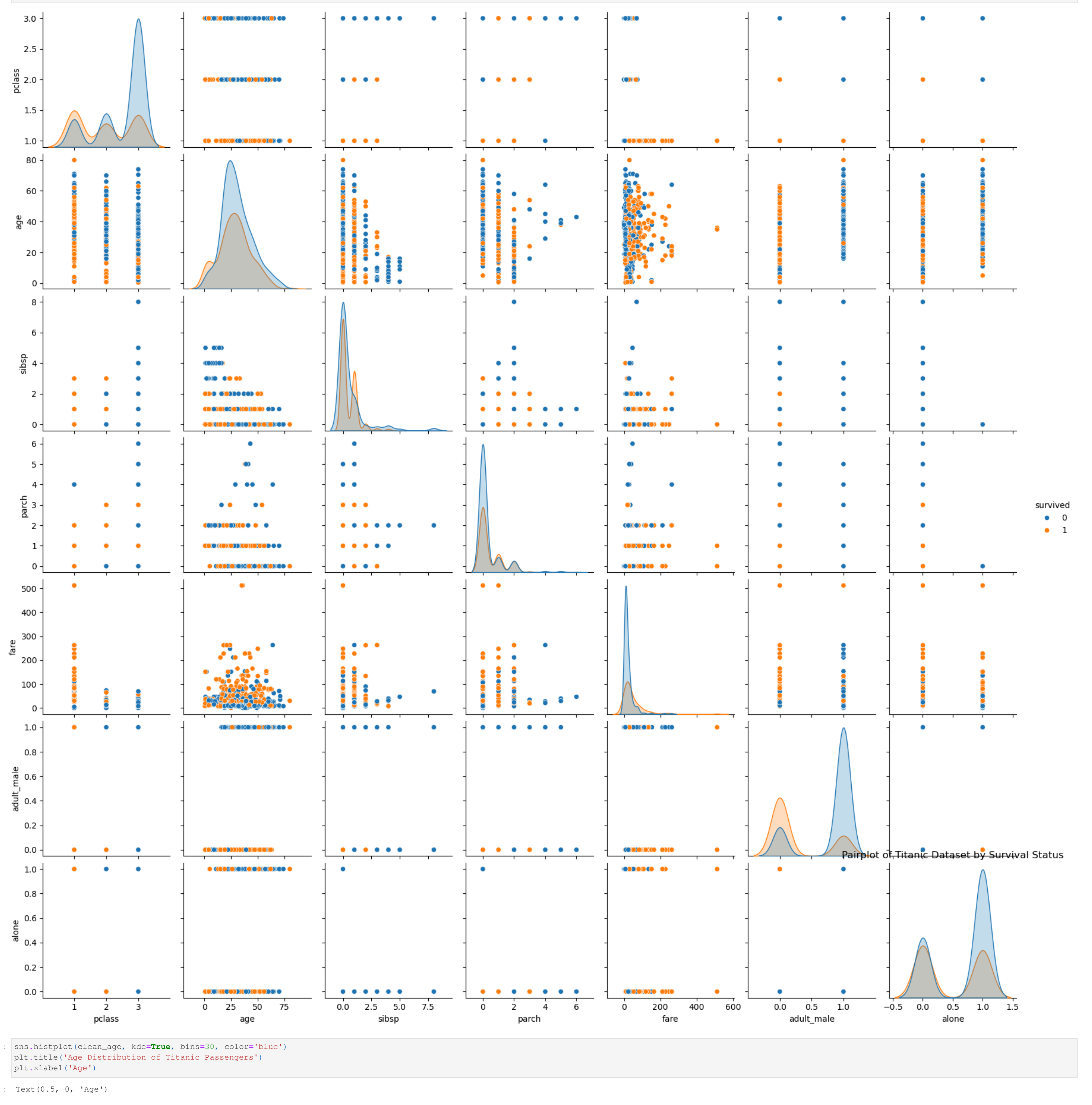
In [29]:
```python
sns.barplot(titanic_dataset, x='survived', y='fare', hue='class', palette='Set3')
plt.title('Average Fare of Survivors vs Non-Survivors by Class')
plt.show()
```



In [30]:
```python
sns.countplot(titanic_dataset, x='class', hue='embark_town', palette='Set1')
plt.title('Passenger Count by Class and Embarkation Town')
plt.show()
```

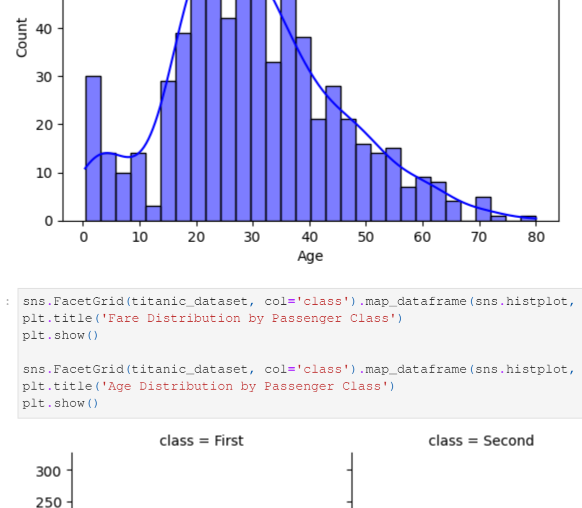


In [31]:
```python
sns.pairplot(titanic_dataset, hue='survived', diag_kind='kde')
plt.title('Pairplot of Titanic Dataset by Survival Status')
plt.show()
```

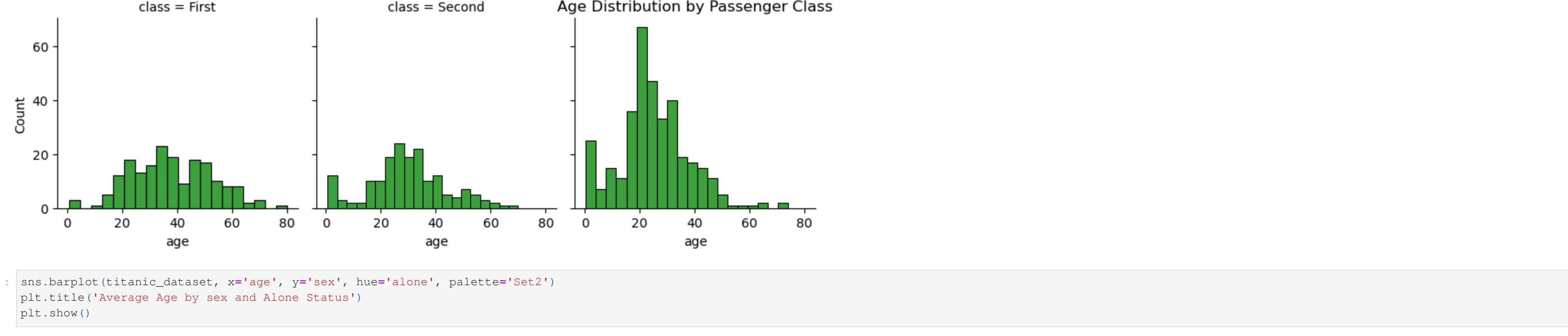


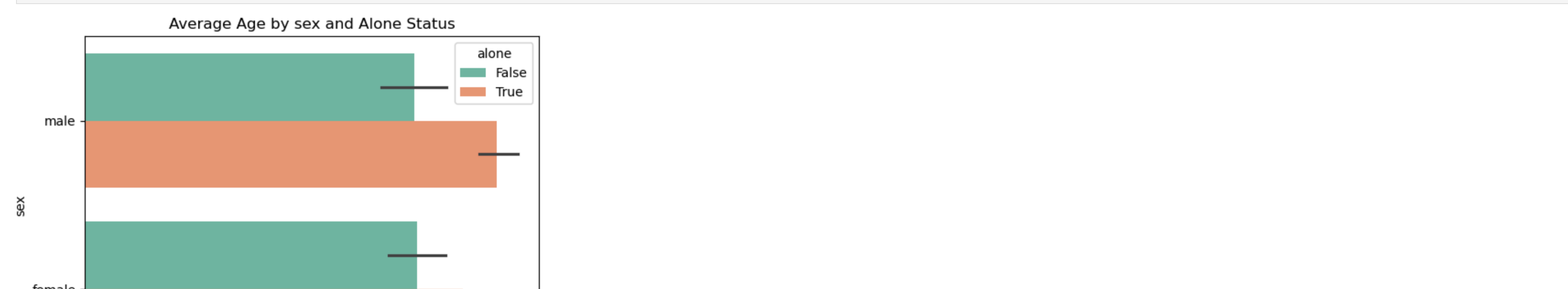
In [32]:
```python
sns.histplot(clean_age, kde=True, bins=30, color='blue')
plt.title('Age Distribution of Titanic Passengers')
plt.xlabel('Age')
```

Out[32]: `Text(0.5, 0, 'Age')`

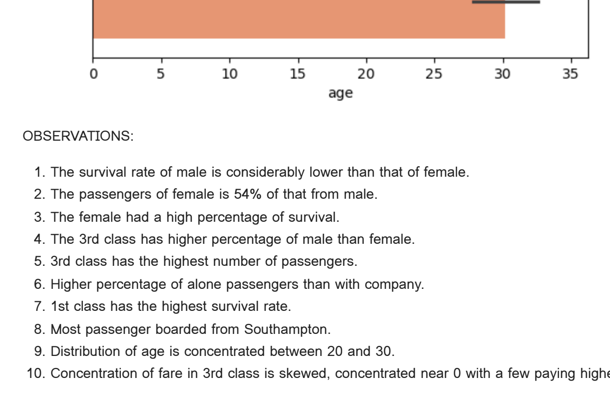


In [33]:
```python
sns.FacetGrid(titanic_dataset, col='class').map_dataframe(sns.histplot, x='fare', bins=20, color='green').add_legend()
plt.title('Fare Distribution by Passenger Class')
plt.show()

sns.FacetGrid(titanic_dataset, col='class').map_dataframe(sns.histplot, x='age', bins=20, color='green').add_legend()
plt.title('Age Distribution by Passenger Class')
plt.show()
```





In [34]:
```python
sns.barplot(titanic_dataset, x='age', y='sex', hue='alone', palette='Set2')
plt.title('Average Age by sex and Alone Status')
plt.show()
```



OBSERVATIONS:

1. The survival rate of male is considerably lower than that of female.
2. The passengers of female is 54% of that from male.
3. The female had a high percentage of survival.
4. The 3rd class has higher percentage of male than female.
5. 3rd class has the highest number of passengers.
6. Higher percentage of alone passengers than with company.
7. 1st class has the highest survival rate.
8. Most passenger boarded from Southampton.
9. Distribution of age is concentrated between 20 and 30.
10. Concentration of fare in 3rd class is skewed, concentrated near 0 with a few paying higher.

11. 3rd class distribution in age is concentrated near 20.

Visual References: -sns.countplot(x='sex', hue='survived', data=titanic) -sns.histplot(data=titanic, x='fare', hue='class', bins=20) -sns.boxplot(x='class', y='age', data=titanic)

INSIGHTS:

1. The percentage of men and women had a huge disparity likely because there are workers who is mostly male in the 3rd class that also has the highest number of passengers.
2. Upon investigation, the ship followed the 'women and children first safety protocol' which is why females had a higher percentage of survival rate.
3. Travelers who is age of 30 to 40 has a higher percentage of 'alone' status reasoning that young is often accompanied by adults.
4. 1st class has the most survival rates because they are near the rescue boats mentioned meanwhile 3rd class has the most casualty because the initial impact of the iceberg was from the third class and with little to no access in rescue boats, they didn't survive.
5. Southampton's dominance aligns with its role as Titanic's departure port, where most emigrants boarded. Factoring that, most of the victims has boarded from Southampton.
6. Third class being Young workforce

Visual References: -sns.barplot(x='class', y='survived', hue='sex', data=titanic) -sns.histplot(x='age', hue='alone', data=titanic, bins=20) -sns.countplot(x='embark_town', hue='class', data=titanic)

SUMMARY: Most of victims boarded from Southampton and would be considered a "young workforce" departing to America. 1st class survived because of easier access in lifeboats This analysis demonstrates how data visualization and simple statistics can uncover real-world social dynamics hidden within historical datasets. By combining Pandas, Seaborn, and NumPy, I learned to: Interpret categorical and numerical data relationships. Understand how patterns in age, fare, and class translate into real human stories. Use data storytelling to make findings more meaningful than numbers alone.